GEONOVA: Scalable Multi-Temporal Satellite Analytics for Environmental Change Detection Using Distributed Data Processing

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Abstract

Manuscript status: protocol-stage draft. No experiment described here has been run, and no measured result appears anywhere in this document. Environmental change analysis from satellite imagery increasingly yields large tabular indicator datasets, yet the computational cost and numerical reproducibility of processing them are seldom reported next to the environmental findings. This study specifies GEONOVA, a workflow that derives the Normalized Difference Vegetation Index (NDVI), the McFeeters Normalized Difference Water Index (NDWI) and the Normalized Difference Built-up Index (NDBI) from Copernicus Sentinel-2 Level-2A surface reflectance for Ahmedabad, Surat and Pune over a proposed 2021–2025 period, subject to data availability. Raster preprocessing is done with Rasterio and NumPy; an identical tabular analytical dataset is then processed by a Pandas baseline and by PySpark with Spark SQL in local mode on a single Windows 11 machine. Four experiments are defined: regional indicator analysis (RQ1), runtime, throughput and memory scaling (RQ2), numerical agreement of the two implementations (RQ3), and sensitivity of regional estimates to cloud masking, missing observations and seasonal window (RQ4). Speedup is not assumed in either direction, and local Spark execution is not treated as equivalent to a multi-node cluster. Spectral-index differences are interpreted as indicators, not as verified land-cover change. Results, discussion and conclusions will be written only from measured outputs.

Keywords: Sentinel-2; environmental change detection; NDVI; NDWI; NDBI; Apache Spark; PySpark; Pandas; benchmarking; Parquet; reproducibility

1. Introduction

The Copernicus Sentinel-2 mission carries a Multi-Spectral Instrument (MSI) that samples 13 spectral bands at 10 m, 20 m and 60 m, with a 290 km swath and a nominal five-day revisit at the equator when two satellites operate together [6], [24]. Level-2A (L2A)

products supply surface reflectance and a Scene Classification Layer (SCL) [23], which makes the archive suitable for repeated environmental indicators such as NDVI [1], NDWI [2] and NDBI [5].

Two questions are usually treated separately. The first is environmental: how do vegetation, water-related and built-up spectral indicators change across places, years and seasons? The second is computational: when indicator extraction produces millions of table rows, which processing engine is appropriate, and do two engines return the same numbers? This study evaluates both in one protocol for a student-scale system, GEONOVA.

The computational comparison is Pandas [15] versus Apache Spark [13], [14] used through PySpark and Spark SQL on one Windows 11 computer. Generic single-machine benchmarks report that the better choice depends on data size and operation type: Pandas is competitive on small data, while Spark's optimizer and spill-to-disk behaviour help on very large data [16]. Whether that pattern holds for satellite-derived index tables, and whether the outputs agree numerically, is not assumed here.

2. Research Motivation and Problem Statement

Three problems motivate the work.

1. **Indicator ambiguity.** NDVI, NDWI and NDBI are spectral ratios. They are not direct measurements of biomass, water extent or built-up area, and two different indices share the name NDWI [2], [3]. A change in an index value therefore needs an explicit interpretation rule and an explicit formula.
2. **Data-quality sensitivity.** Clouds, cloud shadows, missing observations and seasonal phenology can alter regional estimates. Reported accuracies of operational cloud masks differ between studies and regions [8], [9], [10], so mask choice is a modelling decision that should be tested, not assumed.
3. **Unreported computational and numerical behaviour.** Engineering choices such as Pandas or Spark, null handling and aggregation order can change runtime, memory and even the last digits of results. These are rarely documented together with the environmental result.

Problem statement. Given Sentinel-2 L2A observations for three Indian urban regions, quantify multi-year changes in three spectral indicators under explicit quality rules, and measure how equivalent Pandas and PySpark implementations of the same analytical

pipeline compare in runtime, throughput, memory and numerical agreement as input size grows.

3. Related Work

Verification scope. Every entry below was located through web search and its title, venue and abstract or official page were seen in this session. Full texts were not all read. Where a matrix cell needs the full text, it says *verify in full text*. The search was not a systematic review (no Scopus, Web of Science or IEEE Xplore query logs), so the gap in Section 4 stays provisional.

3.1 Spectral indices and their limits

Tucker related red and near-infrared radiance combinations to grass-canopy biomass, leaf water and chlorophyll using in situ spectrometers, and found them sensitive to photosynthetically active vegetation, with accumulated dead vegetation affecting the relationship [1]. McFeeters proposed the green/NIR water index [2]. Gao proposed a different NDWI from NIR and 1.24 μm reflectance for vegetation liquid water [3]; the two share a name but not a purpose. Xu replaced the NIR band with a middle-infrared band to reduce built-up land noise that, in his analysis, contaminated the green/NIR index [4]. Zha et al. introduced NDBI for automatic urban mapping from Landsat TM [5]. Later papers continue to report that NDBI separates built-up land from bare land poorly [29].

3.2 Sentinel-2 data, atmospheric correction and cloud masks

Drusch et al. describe the mission [6]; ESA's SentiWiki documents products, Sen2Cor Level-2A processing, the SCL classes and processing baselines [23], [24]; Main-Knorn et al. describe Sen2Cor [7]. Baetens et al. validated MAJA, Sen2Cor and Fmask cloud masks against reference masks and reported overall accuracies of about 91%, 84% and 90% respectively [8]. A study over the Amazon reported a different ranking, with Fmask 4 highest [9], and a published comment disputes how one mask was interpreted [10]. The practical lesson is that mask accuracy depends on region and evaluation design. No study over Indian cities was found in this search.

3.3 Distributed and single-machine data processing

Spark builds on resilient distributed datasets [13] and the Spark SQL DataFrame API with the Catalyst optimizer [14]. Mozzillo et al. compared Pandas, PySpark and other dataframe libraries on four real datasets (0.2 to 77 million rows) and TPC-H using laptop, workstation

and server configurations; Pandas was best for small data, while Spark SQL handled the largest data on the laptop configuration, and PySpark was slower on simple operations because of distributed-architecture overhead [16]. Raster-oriented Spark systems exist, including RDPro [17], STARK [19] and a Hadoop and RasterFrames framework [20]; GeoSpark, now Apache Sedona, handles vector spatial data [18]. Google Earth Engine offers a hosted alternative [12]. A 2018 vendor blog also benchmarked Spark against Pandas on one node [28]; it is not peer reviewed and is used only as background.

3.4 Regional context

Previous work on these cities used Landsat. A study of Ahmedabad and Surat (2000–2020) used NDBI and supervised land-cover classification and reported larger built-up growth in Surat than in Ahmedabad [22]. A Pune study used Landsat land-cover for 1992, 2001 and 2013 and described diffusion and coalescence growth patterns [21]. Neither is used here as ground truth.

3.5 Literature-review matrix (Table 2)

Table 2A. Remote-sensing literature.

Authors, year	Objective	Dataset and coverage	Methods	Metrics	Principal findings	Limitations noted by authors	Relevance to GEONOVA
Tucker 1979 [1]	Relate red/IR combinations to canopy variables	In situ spectrometer, grass canopy plots, June, September, October	Regression	Regression significance	Red-IR combinations outperformed green-red; sensitive to green vegetation	Dead-vegetation accumulation effect noted; verify in full text	Basis for reading NDVI as a greenness indicator only
McFeeters 1996 [2]	Delineate open water	Verify in full text	Green/NIR normalized difference	Verify in full text	Defines the green/NIR NDWI	Verify in full text; built-up contamination reported by [4]	Water index used in GEONOVA
Gao 1996 [3]	Vegetation liquid water	Imaging-spectrometer and satellite-type data (verify)	NIR/1.24 μm index	Index comparison with NDVI	Sensitive to canopy water; less atmosphere-sensitive than NDVI	Soil background not fully removed	Name clash; not used here

Authors, year	Objective	Dataset and coverage	Methods	Metrics	Principal findings	Limitations noted by authors	Relevance to GEONOVA
Xu 2006 [4]	Improve open-water enhancement	Landsat-type MIR band; study area to verify	NIR replaced by MIR	Qualitative and quantitative water extraction	Better built-up noise suppression than green/NIR index	Verify in full text	Justifies discussing NDWI limits and SWIR alternatives
Zha et al. 2003 [5]	Map urban areas automatically	Landsat TM; study area to verify	SWIR/NIR normalized difference	Mapping accuracy (verify)	Introduces NDBI	Verify in full text; bare-land confusion reported later [29]	Built-up indicator used in GEONOVA
Drusch et al. 2012 [6]	Describe Sentinel-2 mission	Global land and coastal coverage	Mission design	Not applicable	13-band MSI, 10/20/60 m	Not applicable	Primary data source
Main-Knorn et al. 2017 [7]	Describe Sen2Cor L2A processor	Sentinel-2 L1C to L2A	Atmospheric correction and scene classification	Validation results summarised	L2A BOA reflectance with SCL	Verify in full text	Origin of reflectance and SCL
Baetens et al. 2019 [8]	Validate L2A cloud masks	32 reference masks, 10 sites, different seasons	MAJA, Sen2Cor, Fmask vs reference masks	Overall accuracy	About 91%, 84%, 90% respectively	Verify in full text	Motivates mask-sensitivity test (Exp. C)
Sanchez et al. 2020 [9], comment [10]	Compare cloud detectors in Amazon	Five Amazon areas, more than 8,000 validation points	Fmask 4, MAJA, Sen2Cor, s2cloudless	Overall accuracy	Fmask 4 highest (90%) in that study	A published comment disputes MAJA interpretation [10]	Mask rankings vary by region and evaluation
Olofsson et al. 2014 [11]	Good-practice accuracy and area estimation	General land change	Probability sampling, error matrix, area estimation	Overall, user's and producer's accuracy with confidence intervals	Recommends sampling and response design	Not applicable	Template if reference labels are ever created (Eq. 18–20)

Table 2B. Data-processing and regional literature.

Authors, year	Objective	Dataset and coverage	Methods	Metrics	Principal findings	Limitations noted by authors	Relevance to GEONOVA
Zaharia et al. 2012 [13]	Fault-tolerant in-memory cluster computing	Cluster workloads	RDD abstraction	Performance vs prior systems	Coarse-grained transformations enable fault tolerance	Not a general shared-memory model	Spark foundation

Authors, year	Objective	Dataset and coverage	Methods	Metrics	Principal findings	Limitations noted by authors	Relevance to GEONOVA
Armbrust et al. 2015 [14]	Relational processing in Spark	Not applicable	DataFrame API, Catalyst optimizer	Evaluation in paper (verify)	Integrates relational and procedural processing	Verify in full text	Spark SQL path
McKinney 2010 [15]	Data structures for statistical computing	Not applicable	pandas library	Not applicable	Defines pandas	Not applicable	Baseline
Mozzillo et al. 2025 [16]	Compare dataframe libraries on one machine	4 real datasets, 0.2 to 77 million rows, plus TPC-H	Pandas, PySpark, Polars, others; laptop, workstation, server	Runtime, speedup vs Pandas, memory feasibility	Pandas best on small data; Spark SQL scales best on laptop for largest data; PySpark slower on simple operations	Single machine only; distributed left to future work	Closest benchmark design precedent; not satellite data
Shang et al. [17]	Distributed raster processing on Spark	Large raster datasets	RDPro on Spark	Runtime vs other systems	Authors report large speedups over existing systems	Verify in full text	Shows raster Spark systems exist; GEONOVA does not replace raster tools
Yu et al. 2019 [18]	Spatial data management on Spark	Real spatial datasets	GeoSpark	Range, kNN, join runtime	Authors report speedups over Hadoop-based and Spark-based systems	Verify in full text	Vector spatial option for later
Hagedorn et al. 2019 [19]	Raster and vector processing on Spark	Demonstration	STARK extension	Demonstration	Combines raster and vector operators	Demonstration paper	Related raster-on-Spark work
Hadoop and RasterFrames framework [20]	Satellite imagery processing	Surface-water use case	HDFS plus RasterFrames	Verify in full text	Proposes a framework	Requires HDFS; authors to verify	Contrast: GEONOVA avoids HDFS
Ahmedabad and Surat study [22]	Built-up and land-cover dynamics 2000–2020	Landsat, two cities	NDBI plus supervised classification	Built-up change percentages	Surat grew more than Ahmedabad in that study	Authors to verify	Regional context, Landsat not Sentinel-2
Kantakumar et al. [21]	Urban expansion in Pune	Landsat 1992, 2001, 2013	Land-cover and urban-growth metrics	Expansion metrics	Diffusion-coalescence pattern; faster growth after 2001	Verify in full text	Regional context

3.6 What is established and what remains open

- **Established:** index definitions and their caveats [1]–[5]; Sentinel-2 L2A product structure [6], [7], [23], [24]; mask accuracy varies by region [8]–[10]; single-machine Pandas-versus-Spark behaviour on generic tabular data [16]; Spark-based raster and spatial systems exist [17]–[20].

- **Not found in this search:** a study that pairs Sentinel-2-derived index tables for Indian urban regions with a Pandas-versus-Spark benchmark that also checks numerical agreement and the sensitivity of regional estimates to mask and season choices.

4. Research Gap and Contributions

Provisional research gap. Existing work supplies the index definitions [1]–[5], the Sentinel-2 L2A product [6], [7], [23], [24], region- and method-dependent cloud-mask accuracies [8]–[10], generic single-machine Pandas-versus-Spark benchmarks [16], and Spark-based raster or spatial systems [17]–[20]. In the sources found here, these threads are not joined. Specifically, three things appear insufficiently evaluated together:

1. How equivalent Pandas and PySpark pipelines behave, in runtime, throughput, memory and **numerical agreement**, on tables derived from Sentinel-2 L2A indices rather than generic business or Kaggle data.
2. How much regional index estimates for Indian urban regions move when cloud-mask rules, valid-observation requirements and seasonal windows are varied under a fixed, documented protocol.
3. How these two evaluations can be reported in one reproducible protocol with explicit tolerances and explicit separation of real from synthetic data.

This gap is **provisional**. It rests on a non-systematic web search, and it does not claim that no related study exists. Before submission, a systematic search (Scopus, Web of Science, IEEE Xplore, Google Scholar) with logged queries must confirm or narrow it.

Contributions. Contributions will be listed in the final paper only when implemented and supported by completed experiments. At this stage none is claimed as achieved. The intended contributions and the evidence each needs are:

ID	Intended contribution	Evidence required before it is claimed	Status
C1	A reproducible protocol for Sentinel-2 L2A index tables with explicit validity rules	Released scripts, logged parameters, successful rerun	Pending
C2	A measured Pandas-versus-PySpark comparison on index tables, with numerical-agreement checks	Experiments B and D completed with repeated runs	Pending

ID	Intended contribution	Evidence required before it is claimed	Status
C3	A measured sensitivity analysis of regional index estimates to mask, coverage and season	Experiment C completed	Pending
C4	Regional indicator time series for Ahmedabad, Surat and Pune	Experiment A completed with coverage reported	Pending

5. Research Questions and Objectives

Objectives. O1: derive NDVI, NDWI and NDBI under fixed rules. O2: compare Pandas and PySpark on equivalent inputs and logic. O3: test numerical agreement against pre-specified tolerances. O4: quantify sensitivity to data quality. O5: document everything needed to rerun.

Table: mapping of research questions to experiments.

RQ	Question	Experiment	Dataset	Primary metrics	Output to be produced
RQ1	How do vegetation, water-related and built-up indicators vary across regions and years?	A	Real Sentinel-2 L2A composites, 3 regions, 2021–2025	Regional mean per index, valid-pixel coverage, Δ per window pair (Eq. 8–10)	Table 6, Figures 3–5
RQ2	How does performance change with dataset size for Pandas and PySpark?	B	Records sampled from real tables; synthetic scale-up clearly labelled	Runtime, throughput, peak memory, speedup (Eq. 11–14)	Table 4, Figures 6–8
RQ3	Are Pandas and PySpark results numerically consistent?	D	Same inputs as B	MAE, MaxAE, tolerance pass rate (Eq. 15–17)	Table 5, Figure 9
RQ4	How do clouds, missing data and season affect change estimates?	C	Real composites under controlled mask, coverage and window variants	Shift in regional mean and Δ, coverage change	Table 7, Figure 10

6. Dataset Description

6.1 Product specification (verified from official documentation)

The primary dataset is Copernicus Sentinel-2 Level-2A surface reflectance, obtained from the Copernicus Data Space Ecosystem (https://dataspace.copernicus.eu/). The access date, access method and licence text must be recorded at download time.

Property	Value	Source
Instrument	MSI, 13 bands, 290 km swath	[24]
Revisit	5 days at the equator with two satellites; 10 days per satellite	[24]
Operating units	Sentinel-2B and 2C nominal; 2C replaced 2A in operation on 21 January 2025; 2A also in an extension campaign from March 2025	[24]
Native resolution	10 m: B2, B3, B4, B8. 20 m: B5, B6, B7, B8A, B11, B12. 60 m: B1, B9, B10	[24]
Bands used here	B3 green (about 560 nm, 10 m); B4 red (about 665 nm, 10 m); B8 NIR (about 833 nm, 10 m); B11 SWIR (about 1610 nm, 20 m); SCL (20 m)	[23], [24]
Product level	L2A: surface reflectance from Sen2Cor plus Scene Classification Layer; systematic global L2A production since 13 December 2018	[23], [24]
Radiometric offset	Since Processing Baseline (PB) 04.00 (operational 25 January 2022), digital numbers carry a band-dependent offset; ESA's Collection-1 reprocessing applied consistent baselines to the archive from 2015 (PB 05.00 for 4 July 2015 to 31 December 2021; PB 05.10 for 1 January 2022 to 13 December 2023)	[23]
Reflectance conversion	reflectance = (DN + BOA_ADD_OFFSET) / quantification value; for PB 04.00, 05.00, 05.09, 05.10 and 05.11 the offset is −1000 and the quantification value 10,000; both must still be read from product metadata	[25]

Property	Value	Source
NoData	DN = 0	[25]
Negative reflectance	Reflectance can be encoded down to about −0.1; such pixels are not flagged in quality masks	[30]
Scene classification dilation	Since Sen2Cor 2.10 (PB 04.00 or later): clouds +80 m, cloud shadows +40 m, snow +20 m, with exceptions	[23]

Because the offset depends on the processing baseline recorded in each product and not on acquisition date, the pipeline must read `PROCESSING_BASELINE`, `BOA_ADD_OFFSET` and `BOA_QUANTIFICATION_VALUE` from every product's metadata and fail loudly if they are missing. Third-party platforms may already apply the offset, which must not be applied twice [27]. Pre- and post-2022 data are not safely comparable without this check [27].

Scene Classification Layer values used for masking [23]:

SCL value	Class	Primary-analysis treatment
0	NO_DATA	Exclude
1	SATURATED_OR_DEFECTIVE	Exclude
2	CAST_SHADOWS	Exclude
3	CLOUD_SHADOWS	Exclude
4	VEGETATION	Include
5	NOT_VEGETATED	Include
6	WATER	Include
7	UNCLASSIFIED	Exclude in primary; include in sensitivity variant
8	CLOUD_MEDIUM_PROBABILITY	Exclude
9	CLOUD_HIGH_PROBABILITY	Exclude
10	THIN_CIRRUS	Exclude
11	SNOW or ICE	Exclude

Note that SCL value 2 was named DARK_FEATURES before the Collection-1 baseline 05.11 renamed it CAST_SHADOWS [23]; the pipeline must key on the numeric value.

6.2 Study regions and period

Provisional regions are Ahmedabad, Surat and Pune; the period is 2021–2025, subject to cloud conditions, availability and feasible download volume. All three cities lie in longitude 72°E–78°E, which corresponds to UTM zone 43N (EPSG:32643) in the WGS 84 UTM scheme; this is an expectation to confirm from each product's metadata, not an observed fact. Tile identifiers, scene counts, acquisition dates and valid-coverage statistics **must be obtained from the actual downloads**; none is stated here.

Table 1. Dataset characteristics and study regions (to be completed from downloaded metadata).

Region	AOI boundary source and licence	EPSG (confirm)	MGRS tile(s)	Products downloaded	Products after QC	Observations per window (min / median / max)	Median valid-pixel coverage
Ahmedabad	Pending	Pending	Pending	Pending	Pending	Pending	Pending
Surat	Pending	Pending	Pending	Pending	Pending	Pending	Pending
Pune	Pending	Pending	Pending	Pending	Pending	Pending	Pending

6.3 Preprocessing protocol

1. **Acquisition and metadata validation.** For each product record product name, sensing time, platform (2A, 2B, 2C), processing baseline, tile, footprint and file checksum. Reject products with missing or inconsistent offset metadata.
2. **Reflectance scaling.** Apply Eq. 1 per band using metadata values.
3. **Cloud and shadow masking.** Use SCL values in the table above. Variants for Experiment C are defined in Section 10.
4. **Band alignment and resampling.** *Proposed decision D1, for supervisor review:* analyse on the native 20 m grid. B3, B4 and B8 are block-averaged from 10 m to 20 m (2×2 mean of valid pixels); B11 and SCL are used at native 20 m. This avoids interpolating SWIR reflectance into a finer grid. The alternative, upsampling B11 to 10 m by nearest neighbour, is run as a sensitivity variant.
5. **Clipping.** Clip to the AOI polygon in the product CRS; record AOI area and pixel counts.
6. **Temporal compositing.** Use identical month-day windows in every year. Candidate windows are a winter window (November–February) and a pre-monsoon window (March–May), to be finalised only after counting valid observations per window, because cloudiness in the monsoon season is expected to limit coverage but must be verified from the data. For each pixel compute the index per valid observation, then take the per-pixel median across observations. A pixel is kept only if it has at least k_min valid observations (proposal: k_min = 3, to be justified by observed counts).
7. **Missing-data handling.** Never fill or interpolate in the primary analysis. Missing pixels stay missing and are counted.
8. **Quality checks.** Report the distribution of processing baselines; the fraction of pixels with negative reflectance before flooring; the count of pixels dropped by each rule; the geotransform equality of all products in a tile; and the per-window valid-pixel coverage.
9. **Provenance.** Write a JSON manifest per run containing input checksums, parameters, library versions and the git commit of the code.

Atmospheric correction is taken as delivered by Sen2Cor L2A [7], [23]; this study does not re-correct. Landsat Collection 2 is an optional extension and is excluded from the main experiment because cross-sensor comparability is not addressed here.

7. Proposed Methodology

7.1 Two stages, kept separate

Stage 1, raster preprocessing, is done with Rasterio and NumPy: reading JP2 or GeoTIFF bands, applying Eq. 1, masking with SCL, block-averaging to the analysis grid, clipping, compositing, and computing reference index rasters for the maps in Experiment A. Spark is not used for these steps. **Stage 2, tabular analytics,** starts from a Parquet table exported from Stage 1. Pandas and PySpark both read that table and run the same analytical operations. PySpark therefore does not replace raster tooling here. It is evaluated only on tile-level or pixel-level records.

7.2 Roles of the tools

Tool	Role in GEONOVA	Not used for
Rasterio, NumPy	Raster I/O, reflectance scaling, masking, clipping, reference index rasters, composites	Distributed execution
Pandas [15]	Single-process baseline implementation of the analytical pipeline	Out-of-memory data
Apache Spark, PySpark, Spark SQL [13], [14]	Second implementation of the same pipeline in local mode on one machine	Multi-node claims; raster geometry
Parquet	Columnar exchange format for the common analytical dataset and results	Raster storage
PostgreSQL/PostGIS	Storage of region geometries and final results for the dashboard	Benchmarked computation

Tool	Role in GEONOVA	Not used for
FastAPI, React	Serving and displaying stored results	Any measured computation

HDFS and Hadoop are not required. If data exceed local disk or memory, an object store or cloud virtual machine is a practical alternative, but is optional and would change the hardware description.

7.3 Common analytical dataset and pipeline operations

The Parquet table has one row per valid-or-invalid cell observation with columns `region`, `window_id`, `year`, `tile_id`, `cell_id`, `obs_date`, `platform`, `proc_baseline`, `b3`, `b4`, `b8`, `b11` (float64 reflectance after Eq. 1 and block averaging), `scl` (int), and reference columns `ndvi_ref`, `ndwi_ref`, `ndbi_ref` computed by the NumPy stage. Both engines recompute the indices from the band columns so that index calculation is part of the benchmark, and the NumPy reference columns serve as a third check.

Operations, in order, identical in both engines:

1. **P1** Read Parquet.
2. **P2** Filter rows by SCL set and reflectance validity (Section 8).
3. **P3** Compute NDVI, NDWI, NDBI with the denominator rule.
4. **P4** Aggregate by `region`, `window_id`: valid count, mean, sample standard deviation, coverage.
5. **P5** Per-cell median across observations in a window (exact median).
6. **P6** Join two windows per cell; compute ΔI and the paired mean change.
7. **P7** Write result tables to Parquet.

7.4 Rules that make the two engines equivalent

- Same input file, same column types: float64 in both engines.
- **Missing values.** Use SQL null in both. In Pandas, NaN is skipped by `mean`; in Spark a NaN is a value and propagates through `avg`. The pipeline must convert NaN to null (or drop it) before aggregation in both engines and assert that no NaN remains.
- **Division by zero and ANSI mode.** The denominator rule in Eq. 3 removes zero denominators before division. Record the value of `spark.sql.ansi.enabled`, because arithmetic error behaviour depends on it and on the Spark version.

- **Standard deviation.** Use the sample standard deviation (n − 1) in both engines.
- **Median.** Use an exact median in both. Spark's approximate-quantile functions are not equivalent to an exact Pandas median [16].
- **Ordering.** Sort outputs by key before comparison. Floating-point summation order can differ between engines, so aggregates are compared under tolerances (Eq. 17), not for bitwise identity.
- **Execution.** Spark is lazy, so every timed run ends with a write action (P7). Caching is off in the primary measurement; a cached variant is reported separately.

7.5 What must be recorded about the environment

Hardware: CPU model, physical cores and threads, RAM, storage type, Windows 11 build, power plan, other running load. Software: Python, pandas, NumPy, PyArrow, Rasterio, PySpark, Spark and Java versions. Spark configuration: master (`local[*]` or `local[n]`), driver memory, `spark.sql.shuffle.partitions` , `spark.sql.adaptive.enabled` , `spark.sql.ansi.enabled` , default parallelism, Parquet compression codec, session time zone. Local Spark runs the driver and executors in one JVM on one machine; it exercises parallelism across cores but not network shuffle or fault tolerance, so it is **not** a distributed cluster result [13], [26]. The Windows setup details for Spark (Java version, any Hadoop-compatibility files required for local file output) must be taken from the documentation of the chosen Spark version and recorded.

8. Mathematical Formulation

8.1 Reflectance and validity

For band b with digital number DN, offset A_b = BOA_ADD_OFFSET and quantification value Q_b = BOA_QUANTIFICATION_VALUE, read from metadata [23], [25]:

```
\rho_b(x,y,t) = \frac{\mathrm{DN}_b(x,y,t) + A_b}{Q_b} \tag{1}
```

A pixel is valid, v(x,y,t) = 1, only if every one of B3, B4, B8, B11 has DN > 0 (DN = 0 is NoData), each reflectance lies in [ρ_min, ρ_max], and SCL belongs to the valid set 🆖 = {4, 5, 6} (primary). Here ρ_min = −0.1 is the lower bound ESA's offset is designed to encode [30]; ρ_max must be fixed from the observed reflectance distribution before analysis and reported. Small negative reflectances over dark surfaces are floored at zero, and the share of floored pixels is reported:

$$\tilde{\rho}_b = \max\left(\rho_b,\, 0\right) \quad \text{for } v=1 \tag{2}$$

8.2 Normalized difference indices

With a pre-specified denominator threshold ε (candidate 10^{-4} in reflectance units, to be checked against the data and reported), the generic index for band pair (a, c) is

$$I_{a,c}(x,y,t) = \begin{cases} \dfrac{\tilde{\rho}_a - \tilde{\rho}_c}{\tilde{\rho}_a + \tilde{\rho}_c}, & v=1 \ \text{and} \ \tilde{\rho}_a + \tilde{\rho}_c > \varepsilon \\[4pt] \text{undefined (masked)}, & \text{otherwise} \end{cases} \tag{3}$$

Because of Eq. 2, defined values lie in [−1, 1].

NDVI (Eq. 4) uses NIR and red. It is a spectral vegetation indicator; it is not a direct measurement of biomass and does not by itself prove vegetation loss [1].

$$\mathrm{NDVI} = \frac{\tilde{\rho}_{B8} - \tilde{\rho}_{B4}}{\tilde{\rho}_{B8} + \tilde{\rho}_{B4}} \tag{4}$$

NDWI, McFeeters form (Eq. 5) uses green and NIR and is intended to highlight open water [2]. It is not the Gao NDWI (NIR and SWIR, vegetation liquid water) [3]. SWIR-based water indices such as the modified NDWI [4] exist and may be added as a sensitivity analysis. Built-up surfaces can raise the green/NIR index and cause confusion with water [4].

$$\mathrm{NDWI} = \frac{\tilde{\rho}_{B3} - \tilde{\rho}_{B8}}{\tilde{\rho}_{B3} + \tilde{\rho}_{B8}} \tag{5}$$

NDBI (Eq. 6) uses SWIR (B11) and NIR (B8) and responds to surfaces with high SWIR relative to NIR [5]. Bare soil and dry open land can also produce high values, so NDBI alone does not identify built-up area [29]. B8 is block-averaged to the 20 m grid of B11 (decision D1).

$$\mathrm{NDBI} = \frac{\tilde{\rho}_{B11} - \tilde{\rho}_{B8}}{\tilde{\rho}_{B11} + \tilde{\rho}_{B8}} \tag{6}$$

8.3 Temporal change and regional aggregation

$$\Delta I(x,y) = I_{t_2}(x,y) - I_{t_1}(x,y), \quad \text{defined only if both are defined} \tag{7}$$

$$\mathrm{Mean}_I(R,t) = \frac{1}{N_{R,t}} \sum_{(x,y)\in R} v(x,y,t)\, I(x,y,t), \qquad N_{R,t} = \sum_{(x,y)\in R} v(x,y,t) \tag{8}$$

$$C(R,t) = \frac{N_{R,t}}{N_R} \tag{9}$$

If $C(R,t_1)$ differs from $C(R,t_2)$, a difference of regional means can reflect which pixels were observed and not a change in the surface. The paired estimator uses only pixels valid at both times, V_{12}:

$$\overline{\Delta I}(R) = \frac{1}{|V_{12}|} \sum_{(x,y)\in V_{12}} \Delta I(x,y) \tag{10}$$

Both the difference of means and Eq. 10 are reported with the coverage from Eq. 9.

Interpretation of ΔI. Signs are not equivalent to land-cover transitions.

Index	Δ > 0 may indicate	Δ < 0 may indicate	Other explanations to check
NDVI	Greener surface in the window	Less green surface	Crop calendar, rainfall timing, window mismatch, residual cloud or haze, platform change
NDWI (green/NIR)	More water or wetter, or less NIR-bright vegetation	Less water or more vegetation	Built-up and shadow confusion [4], turbidity, season
NDBI	More SWIR-bright surface such as built-up or bare soil	More vegetation or moisture	Bare soil and fallow land [29], soil moisture, season
Any, near 0	No net change, or opposing changes cancelling		Coverage differences, compositing choice

8.4 Performance metrics

$$\mathrm{Speedup} = \frac{T_{\mathrm{Pandas}}}{T_{\mathrm{Spark}}} \tag{11}$$

$$\mathrm{Throughput} = \frac{N_{\mathrm{processed}}}{T_{\mathrm{elapsed}}} \tag{12}$$

For n repeated timed runs:

$$\overline{T} = \frac{1}{n}\sum_{i=1}^{n} T_i \tag{13}$$

$$\mathrm{SD}_T = \sqrt{\frac{\sum_{i=1}^{n} \left(T_i - \overline{T}\right)^2}{n-1}} \tag{14}$$

Speedup is computed from mean runtimes, with the run-to-run SD reported. Values below 1 mean Pandas was faster. Spark startup, JVM warm-up, plan compilation and file-cache effects inflate first-run time, so one warm-up run is discarded and startup is also measured separately.

8.5 Numerical consistency

For matching outputs y_i from the two engines over a comparison population of N values (all rows of each result table, after sorting by key):

$$\mathrm{MAE} = \frac{1}{N}\sum_{i=1}^{N}\left|y_i^{\mathrm{Pandas}} - y_i^{\mathrm{Spark}}\right| \tag{15}$$

$$\mathrm{MaxAE} = \max_{i}\left|y_i^{\mathrm{Pandas}} - y_i^{\mathrm{Spark}}\right| \tag{16}$$

Tolerances are fixed before any comparison (proposal: per-cell index values $a_{tol} = 10^{-12}$, $r_{tol} = 0$; regional aggregates $a_{tol} = 0$, $r_{tol} = 10^{-9}$; both to be justified in the protocol):

$$\left|y_i^{\mathrm{Pandas}} - y_i^{\mathrm{Spark}}\right| \le a_{\mathrm{tol}} + r_{\mathrm{tol}}\left|y_i^{\mathrm{Pandas}}\right| \quad \forall i \tag{17}$$

Computational equivalence (Eq. 15–17) says the two engines agree. It says nothing about scientific validity of the indices.

8.6 Optional detection metrics

Only if independent reference labels exist for a precisely defined change target (for example, "cell changed from vegetated to built-up between t_1 and t_2" labelled from high-resolution imagery by a sampling design such as [11]). TP is a labelled change predicted as

change; FP is a no-change label predicted as change; FN is a labelled change predicted as no-change.

$$\mathrm{Precision} = \frac{TP}{TP+FP} \tag{18}$$

$$\mathrm{Recall} = \frac{TP}{TP+FN} \tag{19}$$

$$F_1 = \frac{2\,\mathrm{Precision}\,\mathrm{Recall}}{\mathrm{Precision}+\mathrm{Recall}} \tag{20}$$

Spectral-index differences must not be used as their own labels. If no suitable labels are created, these metrics are not reported.

8.7 Notation

Symbol	Meaning	Unit or range
DN_b	Digital number of band b	Integer; 0 is NoData
A_b, Q_b	Offset and quantification value from metadata	−1000 and 10,000 for the baselines listed in [25]
ρ_b, ρ̃_b	Surface reflectance, and its zero-floored form	Dimensionless
v	Validity flag	0 or 1
ε	Denominator threshold	Reflectance units
I	NDVI, NDWI or NDBI	[−1, 1]
ΔI	Index difference between windows	[−2, 2]
R, t	Region and window	
N_{R,t}, C(R,t)	Valid-pixel count and coverage	Count; [0, 1]
T, N_processed	Elapsed time and number of records processed	s; records
a_tol, r_tol	Absolute and relative tolerance	Pre-specified

9. System Architecture

9.1 Figure 1: GEONOVA system architecture

```
flowchart TB
  A["Satellite Data Sources<br/>Copernicus Sentinel-2 L2A, accessed via the
Copernicus Data Space Ecosystem"] --> B["Data Acquisition and Metadata
Validation<br/>Python scripts, checksums, processing-baseline and offset
check"]
  subgraph RASTER["STAGE 1 - Raster preprocessing (Rasterio, NumPy). Not done
in Spark."]
    direction TB
    C1["Cloud and shadow masking (SCL)"] --> C2["Band alignment to 20 m
grid"]
    C2 --> C3["Study-area clipping"]
    C3 --> C4["Temporal compositing"]
  end
  B --> C1
  C4 --> D["Feature Extraction<br/>NDVI, NDWI, NDBI index rasters (NumPy)"]
  D --> E["Common Analytical Dataset<br/>Parquet table: band reflectances,
SCL, reference index columns"]
  subgraph TABULAR["STAGE 2 - Tabular analytics: same input, same rules, same
output schema"]
    direction LR
    F1["A. Pandas baseline"]
    F2["B. PySpark and Spark SQL (local mode)"]
  end
  E --> F1
  E --> F2
  F1 --> G["Performance and Numerical Evaluation<br/>runtime, throughput,
memory, MAE, MaxAE"]
  F2 --> G
  G --> H["Environmental Change Analysis<br/>regional means, change,
coverage, sensitivity"]
  D -. "index rasters for maps (Fig. 4)" .-> H
  H --> I["Parquet Results and Geospatial Storage<br/>Parquet files,
PostgreSQL with PostGIS"]
  I --> J["GEONOVA Dashboard<br/>FastAPI service, React front end"]
```

Caption. GEONOVA architecture. Raster-specific preprocessing (Rasterio, NumPy) is separate from tabular analytics, where Pandas and PySpark with Spark SQL process one shared Parquet table. PySpark is applied to tile-level or pixel-level records only; it does not

replace raster tooling. PostgreSQL with PostGIS stores geometries and final results; FastAPI and React serve and display them. These last two components are not benchmarked.

9.2 Reading the figure

Acquisition validates metadata before any pixel is read, because the reflectance offset depends on the processing baseline [23], [25]. Stage 1 produces cleaned and composited rasters and the index rasters used for maps. The common analytical dataset exports band reflectances and the NumPy reference indices, so both engines recompute indices from bands and the reference acts as a check. Evaluation compares the engines before any environmental result is trusted. The dashed edge shows that maps come from the raster stage while statistics come from the tabular stage.

9.3 Figure 2: research methodology flowchart

```
flowchart TB
  S1["1. Define study areas and time windows"] --> S2["2. Acquire Sentinel-2
L2A observations"]
  S2 --> S3["3. Data-quality checks: metadata, baseline, offset, coverage"]
  S3 --> Q1{"Enough valid observations per window? (k_min and coverage
rule)"}
  Q1 -- "No" --> R1["Widen window, or drop the window or region; log the
decision"]
  R1 --> S1
  Q1 -- "Yes" --> S4["4. Preprocess and align imagery: mask, 20 m grid,
clip"]
  S4 --> S5["5. Generate seasonal composites"]
  S5 --> S6["6. Calculate NDVI, NDWI, NDBI"]
  S6 --> S7["7. Produce common analytical dataset (Parquet)"]
  S7 --> S8["8. Execute Pandas and PySpark implementations"]
  S8 --> S9["9. Run scalability benchmarks: warm-up plus repeated runs"]
  S9 --> S10["10. Compare numerical outputs with pre-specified tolerances"]
  S10 --> Q2{"Within tolerance?"}
  Q2 -- "No" --> R2["Diagnose cause: nulls, ordering, dtype, ANSI mode. Fix
genuine mismatches, report any remaining discrepancy"]
  R2 --> S8
  Q2 -- "Yes" --> S11["11. Evaluate environmental changes and robustness:
Experiments A and C"]
  S11 --> Q3{"Independent reference labels available?"}
  Q3 -- "Yes" --> L1["Compute precision, recall, F1 using a sampling design"]
```

```
  Q3 -- "No" --> L2["Report index differences only; no classification
metrics"]
  L1 --> S12["12. Generate figures and report results"]
  L2 --> S12
```

Caption. Research methodology for GEONOVA. Two decision branches handle insufficient valid observations and missing reference labels; a third handles numerical disagreement between engines. Pre-specified tolerances and thresholds are fixed before steps 8–10 are run.

10. Experimental Design

All parameters marked *pre-specified* must be written into a protocol file and committed before the experiment is run.

Experiment A: multi-region environmental analysis (RQ1)

- **Input:** real Sentinel-2 L2A composites for Ahmedabad, Surat and Pune, 2021–2025, per window (Section 6.3).
- **Procedure:** compute NDVI, NDWI and NDBI (Eq. 4–6); aggregate per region and window (Eq. 8, 9); compute ΔI between chosen window pairs and the paired estimate (Eq. 7, 10).
- **Outputs:** regional indicator time series; seasonal change maps; regional summary tables; valid-observation coverage per region and window; inter-region comparison. Every regional mean is shown with its coverage.
- **Interpretation rule:** changes are described as changes in the indicator. Claims about deforestation, urbanisation, drought or water-body change require independent validation, such as high-resolution imagery review under a sampling design [11].

Experiment B: Pandas versus PySpark performance (RQ2)

- **Input and schema:** the Parquet table in Section 7.3. Target sizes are 100,000, 1 million and 5 million records, adjusted to hardware. Real records are sampled from the Experiment A table with a fixed seed. If real records are insufficient, additional records are produced by seeded bootstrap resampling of real rows and flagged `is_synthetic = true`. Synthetic results are reported in separate rows and never used for environmental findings.
- **Operations:** P1–P7 (Section 7.3).

- **Configurations:** Pandas (single process); Spark `local[*]`; Spark `local[1]`, which separates engine overhead from multi-core parallelism.
- **Warm-up and repeats:** one warm-up run discarded, then n timed runs (proposal n = 10), each in a fresh process for memory measurement; Spark session start-up time is measured and reported separately.
- **Timing boundaries:** T_total runs from the start of P1 to the completion of the P7 write; for Spark the session already exists, and a write action forces execution. The reported speedup uses T_total without start-up; a second speedup that includes start-up is also reported.
- **Memory:** peak resident set size of the Python process and all child processes (the Spark JVM and Python workers), sampled at a fixed interval (proposal 100 ms) by a separate monitor, using a library such as psutil. Also record the configured Spark driver memory.
- **Output validation:** every run's outputs are compared with the first Pandas run under Eq. 17 before its timing is accepted.
- **Hardware and software:** Table 3.

Experiment C: data-quality robustness (RQ4)

One factor changes at a time relative to the primary variant V0. Outputs are the shift in regional mean and in ΔI, the change in coverage, and whether the ranking of regions changes.

Variant	Change relative to V0	Question answered
V0	Primary: SCL values 4, 5, 6; k_min = 3; 20 m grid	Reference
V1	Also include SCL 7 (unclassified)	Effect of the unclassified class
V2	No cloud or shadow mask	Size of contamination effect
V3	Extra exclusion of cells adjacent to masked cells	Effect of stricter masks beyond Sen2Cor dilation [23]
V4	k_min = 1, 3, 5	Effect of minimum observation count
V5	Randomly remove 10%, 25%, 50% of valid observations, repeated with logged seeds	Sensitivity to missing observations

Variant	Change relative to V0	Question answered
V6	Shift one window by ±1 month; compare same-window and different-window pairs	Seasonal mismatch
V7	Upsample B11 to 10 m instead of aggregating to 20 m	Resampling effect

Experiment D: numerical consistency (RQ3)

For identical inputs, compare each output table from the two engines (and the NumPy reference columns) with MAE, MaxAE and tolerance pass rate (Eq. 15–17); report regional aggregate differences; list every discrepancy with its diagnosed cause (null handling, summation order, ANSI mode, type casting). Computational equivalence is not scientific validity.

11. Evaluation Metrics

Metric	Definition	Equation	Experiment
Mean runtime and SD	Mean and sample SD over n timed runs	13, 14	B
Speedup	T_Pandas / T_Spark; below 1 means Pandas faster	11	B
Throughput	Records per second	12	B
Peak memory	Peak RSS of process tree	Section 10	B
MAE, MaxAE, pass rate	Agreement between engines	15–17	D
Regional mean, coverage, paired change	Indicator level and change	8–10	A, C
Precision, recall, F1	Only with independent labels	18–20	Optional

No significance claim is made unless a statistical test is pre-specified; runtimes are reported as mean ± SD with individual runs available.

11.1 Table templates (all values pending)

Table 3. Experimental hardware and software configuration.

Item	Value
CPU model, physical cores, threads	Pending
RAM (GB)	Pending
Storage type	Pending
Windows 11 build, power plan	Pending
Python, pandas, NumPy, PyArrow, Rasterio versions	Pending
Java, Spark, PySpark versions	Pending
Spark master, driver memory, shuffle partitions, AQE, ANSI mode, Parquet codec	Pending
Random seeds	Pending

Table 4. Runtime, throughput and memory comparison.

Records	Data origin	Engine	n runs	Mean T (s)	SD T (s)	Throughput (records/s)	Peak RSS (MB)	Speedup vs Pandas
100,000	Pending	Pandas	Pending	Pending	Pending	Pending	Pending	1 (reference)
100,000	Pending	Spark local[*]	Pending	Pending	Pending	Pending	Pending	Pending
100,000	Pending	Spark local[1]	Pending	Pending	Pending	Pending	Pending	Pending
1,000,000	Pending	Pandas	Pending	Pending	Pending	Pending	Pending	1 (reference)
1,000,000	Pending	Spark local[*]	Pending	Pending	Pending	Pending	Pending	Pending
1,000,000	Pending	Spark local[1]	Pending	Pending	Pending	Pending	Pending	Pending
5,000,000	Pending	Pandas	Pending	Pending	Pending	Pending	Pending	1 (reference)
5,000,000	Pending	Spark local[*]	Pending	Pending	Pending	Pending	Pending	Pending

Records	Data origin	Engine	n runs	Mean T (s)	SD T (s)	Throughput (records/s)	Peak RSS (MB)	Speedup vs Pandas
5,000,000	Pending	Spark local[1]	Pending	Pending	Pending	Pending	Pending	Pending

Table 5. Numerical consistency between Pandas and Spark.

Output table	N values compared	MAE	MaxAE	Tolerance (a_tol, r_tol)	Pass rate	Explanation of any discrepancy
Per-cell indices (P3)	Pending	Pending	Pending	Pending	Pending	Pending
Regional aggregates (P4)	Pending	Pending	Pending	Pending	Pending	Pending
Per-cell medians (P5)	Pending	Pending	Pending	Pending	Pending	Pending
Change table (P6)	Pending	Pending	Pending	Pending	Pending	Pending

Table 6. Environmental change summaries by region and year. Extend with one row per region, index and window pair.

Region	Index	Window pair	Mean at t1	Mean at t2	Paired change (Eq. 10)	Coverage at t1	Coverage at t2	Valid pixels in both
Ahmedabad	NDVI	Pending	Pending	Pending	Pending	Pending	Pending	Pending
Ahmedabad	NDWI	Pending	Pending	Pending	Pending	Pending	Pending	Pending
Ahmedabad	NDBI	Pending	Pending	Pending	Pending	Pending	Pending	Pending
Surat	NDVI	Pending	Pending	Pending	Pending	Pending	Pending	Pending
Surat	NDWI	Pending	Pending	Pending	Pending	Pending	Pending	Pending
Surat	NDBI	Pending	Pending	Pending	Pending	Pending	Pending	Pending
Pune	NDVI	Pending	Pending	Pending	Pending	Pending	Pending	Pending
Pune	NDWI	Pending	Pending	Pending	Pending	Pending	Pending	Pending
Pune	NDBI	Pending	Pending	Pending	Pending	Pending	Pending	Pending

Table 7. Robustness analysis under different data-quality conditions. Repeat per region.

Variant	Region	Index	Regional mean	Shift vs V0	Paired change shift vs V0	Coverage	Region ranking changed?
V0	Pending	Pending	Pending	0 (reference)	0 (reference)	Pending	Reference
V1	Pending	Pending	Pending	Pending	Pending	Pending	Pending
V2	Pending	Pending	Pending	Pending	Pending	Pending	Pending
V3	Pending	Pending	Pending	Pending	Pending	Pending	Pending
V4	Pending	Pending	Pending	Pending	Pending	Pending	Pending
V5	Pending	Pending	Pending	Pending	Pending	Pending	Pending
V6	Pending	Pending	Pending	Pending	Pending	Pending	Pending
V7	Pending	Pending	Pending	Pending	Pending	Pending	Pending

11.2 Figure specifications and captions

Figure	Content	X axis	Y axis or encoding	Uncertainty shown	Caption (to finalise)
1	System architecture	Not applicable	Mermaid in Section 9.1	Not applicable	Done as source
2	Research methodology flowchart	Not applicable	Mermaid in Section 9.3	Not applicable	Done as source
3	Study-area map	Longitude	Latitude; AOI boundaries, tile footprints, scale bar, CRS	Not applicable	Study areas and Sentinel-2 tile footprints
4	Environmental-index maps	Easting	Northing; one panel per region, index and window; shared colour scale; masked cells shown distinctly	Coverage per panel	Index maps per window; masked cells in grey
5	Multi-year regional index trends	Year or window	Regional mean (dimensionless)	Spread across cells; coverage as a secondary panel	Regional means with coverage
6	Runtime versus dataset size	Records (log)	Mean runtime (s, log)	SD error bars; individual runs	Runtime by engine and size

Figure	Content	X axis	Y axis or encoding	Uncertainty shown	Caption (to finalise)
7	Speedup versus dataset size	Records (log)	Speedup; line at 1	Propagated run-to-run spread	Speedup of Spark over Pandas; below 1 favours Pandas
8	Memory consumption comparison	Records (log)	Peak RSS (MB)	Across runs	Peak memory by engine and size
9	Numerical error comparison	Output table	MAE and MaxAE (log)	Not applicable	Pandas versus Spark agreement
10	Sensitivity of estimates	Variant	Shift in regional mean or change	Replicates for V5	Effect of masking, missing data and season

Reproducible plotting script (Figures 6–8). It reads a results file produced by the benchmark; it contains no data.

```
# plot_benchmark.py  -- reads results/benchmark_runs.csv with columns:
# engine, n_records, run_id, seconds, peak_rss_mb, is_synthetic
import pandas as pd
import matplotlib.pyplot as plt

df = pd.read_csv("results/benchmark_runs.csv")
g = (df.groupby(["engine", "n_records", "is_synthetic"])
       .agg(mean_s=("seconds", "mean"), sd_s=("seconds", "std"),
            mean_mb=("peak_rss_mb", "mean"), sd_mb=("peak_rss_mb", "std"))
       .reset_index())

def panel(ax, ycol, sdcol, ylabel):
    for (eng, syn), s in g.groupby(["engine", "is_synthetic"]):
        label = f"{eng}{' (synthetic)' if syn else ''}"
        ax.errorbar(s["n_records"], s[ycol], yerr=s[sdcol], marker="o",
capsize=3, label=label)
    ax.set_xscale("log"); ax.set_xlabel("Records"); ax.set_ylabel(ylabel);
ax.legend()

fig, ax = plt.subplots(figsize=(6, 4)); panel(ax, "mean_s", "sd_s", "Mean
runtime (s)")
ax.set_yscale("log"); fig.tight_layout();
fig.savefig("figures/fig6_runtime.png", dpi=300)
```

```
fig, ax = plt.subplots(figsize=(6, 4)); panel(ax, "mean_mb", "sd_mb", "Peak
RSS (MB)")
fig.tight_layout(); fig.savefig("figures/fig8_memory.png", dpi=300)

# Figure 7 (speedup): divide Pandas mean_s by Spark mean_s per n_records and
draw a line at 1.
```

12. Results

All results are pending. No experiment has been run. The tables in Section 11.1 are blank templates, and no sentence in this section may be read as a finding. Results text will be written only from outputs that exist as files and pass the checks below.

Outputs that must exist before this section is written.

Needed output	Feeds	Produced?
Download manifest: product names, sensing dates, platforms, processing baselines, tiles, checksums, access date	Table 1	No
Per-window valid-pixel coverage for each region	Tables 1, 6	No
Index rasters and composites per region and window	Figures 4, 5	No
Regional means, paired changes, coverage	Table 6, Figure 5	No
Hardware and software inventory with Spark configuration	Table 3	No
Raw benchmark run log (engine, size, run id, seconds, peak RSS, data origin)	Table 4, Figures 6–8	No
Engine-versus-engine comparison files with tolerance results	Table 5, Figure 9	No
Variant outputs V0–V7 for each region	Table 7, Figure 10	No
Reference labels, if any, with sampling design	Eq. 18–20	No

Worked calculations (HYPOTHETICAL numbers for explanation only; not data and not findings). If a pixel had $\tilde{\rho}_{B8} = 0.40$ and $\tilde{\rho}_{B4} = 0.10$, Eq. 4 would give NDVI = (0.40 − 0.10) / (0.40 + 0.10) = 0.6. If a Pandas run took 10 s and a Spark run took 5 s, Eq. 11 would give a speedup of 2; if the times were reversed the speedup would be 0.5, meaning Pandas was faster. These values are not predictions.

12.1 Result subsections to write once data exist

1. Data summary: products, coverage and exclusions (Table 1).
2. Experiment A: indicator levels, changes and coverage by region (Table 6; Figures 3–5).
3. Experiment B: runtime, throughput and memory by size and engine (Table 4; Figures 6–8), with synthetic rows reported separately.
4. Experiment D: numerical agreement and explained discrepancies (Table 5; Figure 9).
5. Experiment C: sensitivity of estimates to each variant (Table 7; Figure 10).

13. Discussion

The discussion will be written from the results. Interpretation rules fixed in advance, so that conclusions do not depend on which outcome occurs:

- **Performance.** Whatever the speedup is, report it with run-to-run SD and the hardware. If Pandas is faster at some or all sizes, report that and discuss start-up, scheduling and serialization overhead using the separately measured start-up time. If Spark is faster at larger sizes, discuss which operations drive it (for example the aggregation and the median) before generalising. Local Spark uses cores on one machine; conclusions about clusters are not drawn [13], [26]. Prior single-machine work found that the better engine depends on size and operation [16], which this study may or may not reproduce on satellite-derived tables.
- **Numerical agreement.** If all outputs pass Eq. 17, state that the implementations are computationally equivalent under the stated tolerances. If any fail, give the diagnosed cause. Neither outcome says the indices are scientifically valid.
- **Environmental estimates.** Describe changes as indicator changes. For each notable change, list the alternative explanations from the interpretation table in Section 8.3 and what evidence would separate them (for example, window alignment, coverage, high-resolution image review). Compare with Landsat-era studies of these cities [21], [22] only qualitatively and with sensor differences stated.

- **Robustness.** Report which variants moved estimates more than the observed between-region or between-year differences, because only those differences are at risk from data-quality choices.
- **Limits on generality.** Three regions, one sensor family and one machine do not support claims about other locations or hardware.

14. Threats to Validity

Threat	Why it matters	Mitigation in this protocol	Residual risk
Cloud and shadow contamination	Can bias index means and changes	SCL masking; variants V1–V3; coverage reported [8], [9]	SCL errors remain; no India-specific mask validation
Seasonal and phenological variation	Window differences can look like change	Identical month-day windows; V6	Interannual weather differences remain
Differences in spatial resolution	10 m and 20 m bands are not co-sampled	Common 20 m grid (D1)	Mixed pixels at 20 m
Resampling effects	Averaging or interpolation changes values	V7 compares both	Choice of method still affects absolute values
Spectral-index limitations	Indices are ratios, not direct measurements [1], [2]	Interpretation table; no land-cover claims without validation	Indicator meaning stays ambiguous
Bare-soil confusion in NDBI	High NDBI over dry open land [29]	Report NDBI with NDVI and caveats	Cannot separate built-up from bare soil without labels
NDWI built-up confusion	Green/NIR index can respond to built-up surfaces [4]	State formula; optional SWIR-based index	Water extent not validated
Incomplete geographic coverage	Tile edges and AOI edges lose pixels	Coverage per region and window	Edge effects

Threat	Why it matters	Mitigation in this protocol	Residual risk
Missing observations	Differences in valid pixels shift means	Paired change (Eq. 10); k_min; V4, V5	Small samples in cloudy windows
Processing-baseline and offset errors	Wrong offset biases all indices, especially over dark surfaces	Read offset from metadata; fail on missing; checksum log [23], [25], [27]	Third-party mirrors may pre-apply offsets
Satellite-unit changes	2A, 2B and 2C have slightly different spectral parameters [24]	Record platform; test platform effect if sample allows	Residual inter-unit differences
Hardware dependence of benchmarks	Results depend on CPU, RAM, storage, Windows	Full hardware table; no claims beyond this machine	Not transferable
Spark initialisation and caching	JVM start-up, warm-up and file cache distort first runs	Warm-up run, separate start-up timing, fresh processes, cached variant separate	OS file cache effects
Fairness of Pandas versus Spark	Different defaults, parallelism and null semantics	Section 7.4 rules; `local[1]` and `local[*]`; equal inputs and outputs	Implementation effort may differ
Synthetic versus real data	Bootstrap-resampled rows share real values and miss real structure	Flag and separate synthetic rows	Scaling behaviour may differ for real larger data
Absence or limits of reference labels	No ground truth for change	No classification metrics without independent labels [11]	Accuracy of change unknown
Generalising from three regions	Cities differ in climate, form and land cover	State scope	No claim for other locations

15. Limitations

- This is a protocol-stage manuscript; no measured results exist yet.
- The literature search was not systematic and some matrix cells require full-text reading, so the research gap is provisional.
- Local Spark on one Windows 11 computer does not test network shuffle, fault tolerance or cluster scaling, and is not a multi-node result.
- Spectral-index changes are not verified land-cover change.
- The analysis uses a 20 m grid, so finer 10 m detail is not used in the primary analysis.
- Landsat is excluded from the main experiment because cross-sensor comparability is not addressed.
- PostGIS, FastAPI and React are described as system components but are not part of the benchmarked computation.

16. Conclusion and Future Work

Conclusion (to be written after results). This draft specifies a reproducible protocol that evaluates satellite-derived environmental indicators and the computational behaviour of two processing engines together. No conclusion about environmental change, Pandas, Spark or numerical agreement can be drawn yet. The conclusion will state only what Experiments A–D measure.

Future work, conditional on results:

- A multi-node or cloud Spark deployment to test the distributed case.
- Independent reference labels with a probability sampling design [11], enabling Eq. 18–20.
- Comparison with other cloud masks (for example Fmask or MAJA) over Indian cities [8].
- Additional engines such as Polars, which prior single-machine work found strong on in-memory data [16].
- Landsat Collection 2 for longer history, once cross-sensor harmonisation is addressed.
- Spark-based raster or spatial systems for the raster stage [17]–[19].

17. References

Verification status. Each entry below was found in this session on a publisher, repository, ESA or official project page showing its title and venue. DOIs and URLs are copied from pages seen then. None has yet been opened to confirm that the full text supports the statement it is cited for; that check is listed in Work Remaining. Entries marked *incomplete* lack author, year or DOI details that must be completed. Style: IEEE numbered.

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Appendix A. Reproducibility Checklist

- [ ] Sentinel-2 product identifiers, sensing dates, platforms, tiles and processing baselines listed in a committed manifest
- [ ] Data access dates and access method recorded; no tokens, passwords or API keys in any file
- [ ] AOI boundary files with source, licence and CRS
- [ ] Band selection (B3, B4, B8, B11, SCL), valid SCL set, ρ_min, ρ_max, ε, k_min, window dates, resampling method (D1)
- [ ] Library and runtime versions: Python, pandas, NumPy, PyArrow, Rasterio, PySpark, Spark, Java
- [ ] Hardware table (Table 3) and Windows 11 build
- [ ] Spark configuration, including ANSI mode, shuffle partitions, driver memory, master
- [ ] Benchmark input sizes, data origin per row (real or synthetic), and sampling procedure
- [ ] Random seeds for sampling, bootstrap and variant V5
- [ ] Timing boundaries, warm-up policy, number of repetitions, memory-sampling interval
- [ ] Evaluation and plotting scripts under version control, with commit hash in each run manifest
- [ ] Output schemas for every Parquet result table
- [ ] Data licensing and attribution text for Copernicus Sentinel data and for AOI boundary data
- [ ] README with the exact commands to rerun every experiment from the manifest

Appendix B. Work Remaining Before Submission

Data and implementation

- [ ] Download Sentinel-2 L2A products; build the manifest; confirm processing baselines and offsets per product
- [ ] Choose and document AOI boundaries; confirm UTM zone and tile IDs from metadata
- [ ] Count valid observations per candidate window; finalise windows, k_min and drop rules; check the expectation that monsoon-season coverage is low
- [ ] Implement and test: raster stage, Parquet export, Pandas pipeline, PySpark pipeline, benchmark harness, memory monitor
- [ ] Unit-test the engine-semantics claims in Section 7.4 (NaN versus null, ANSI mode, standard deviation, exact median) on the chosen versions

- [ ] Settle Windows 11 Spark setup and record it

Pre-specified parameters (fix and commit before running)

- [ ] ρ_max, ε, k_min, floor policy, tolerances a_tol and r_tol, number of timed runs, sampling interval, benchmark sizes

Experiments

- [ ] Experiment A, with coverage for every estimate
- [ ] Experiment B, with synthetic rows separated
- [ ] Experiment C, variants V0–V7
- [ ] Experiment D, with every discrepancy explained
- [ ] Fill Tables 1 and 3–7 and Figures 3–10 from files only

Validation

- [ ] Decide whether independent reference labels will be created; if yes, design the sample [11] before computing Eq. 18–20
- [ ] Review high-resolution imagery for any region where a real-world change (urbanisation, vegetation loss, water change) will be claimed
- [ ] Spot-check SCL masks against imagery in the study regions

Literature and references

- [ ] Run a systematic search with logged queries and update the gap statement or narrow it
- [ ] Read full texts for every cell marked *verify in full text* and complete Table 2
- [ ] Open each reference and confirm it supports the statement citing it; resolve missing DOIs for [1], [4], [6], [16]
- [ ] Complete author, year and DOI fields for [10], [17], [18], [20], [21], [22], [29]
- [ ] Replace support-forum sources [25], [30] with the Product Specification Document or official documentation if possible
- [ ] Add citations for official documentation of Rasterio, Parquet, PostGIS, PySpark and Pandas versions used; none was verified in this session
- [ ] Confirm Copernicus data licence and attribution wording

Manuscript decisions

- [ ] **Title.** Consider replacing the working title with one that does not imply a distributed cluster, for example "Reproducible Sentinel-2 Spectral-Index Analytics and a Pandas-versus-PySpark Benchmark for Three Indian Urban Regions," because local Spark is not

distributed processing. This suggestion is for supervisor decision.

- [ ] Confirm decision D1 (20 m grid), the primary valid SCL set and the floor-at-zero policy
- [ ] Decide whether to add a SWIR-based water index as a sensitivity analysis
- [ ] Choose target venue and its reference style and page limit
- [ ] Supervisor review of design, claims and wording; do not describe the paper as publication-ready until experiments, validation and reference checks are complete
- [ ] Rewrite Abstract, Results, Discussion, Conclusion and the Contributions table from measured outputs only